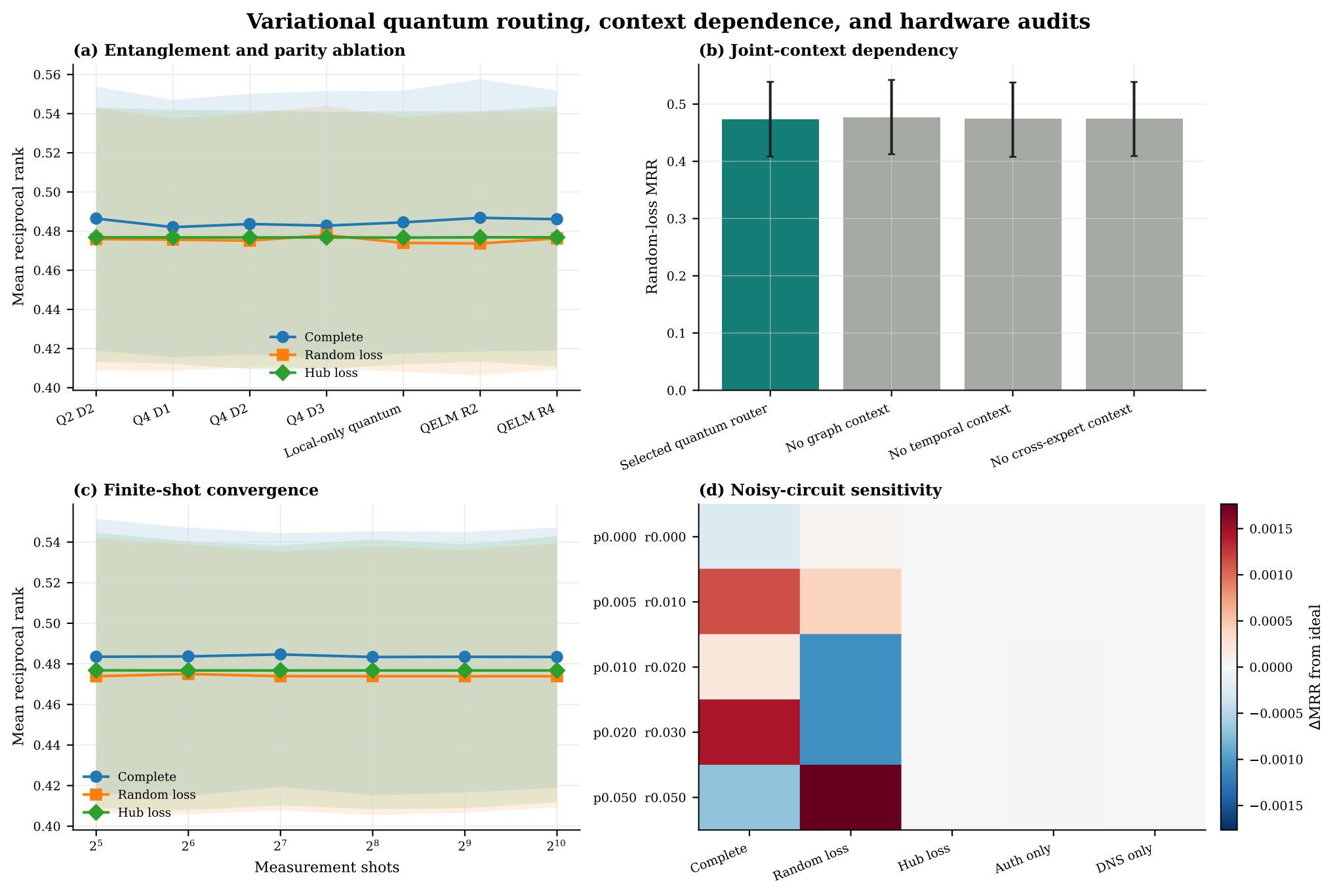
Variational quantum routing, context dependence, and hardware audits
(a) Entanglement and parity ablation
Mean reciprocal rank
0.56
0.54
0.52
0.50
0.48
0.46
0.44
0.42
0.40
Complete
Random loss
Hub loss
Q2 D2
Q4 D1
Q4 D2
Q4 D3
Local-only quantum
QELM R2
QELM R4
(b) Joint-context dependency
Random-loss MRR
0.5
0.4
0.3
0.2
0.1
0.0
Selected quantum router
No graph context
No temporal context
No cross-expert context
(c) Finite-shot convergence
Mean reciprocal rank
0.54
0.52
0.50
0.48
0.46
0.44
0.42
0.40
Complete
Random loss
Hub loss
2^5
2^6
2^7
2^8
2^9
2^{10}
Measurement shots
(d) Noisy-circuit sensitivity
p0.000 r0.000
p0.005 r0.010
p0.010 r0.020
p0.020 r0.030
p0.050 r0.050
Complete
Random loss
Hub loss
Auth only
DNS only
0.0015
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015
ΔMRR from ideal